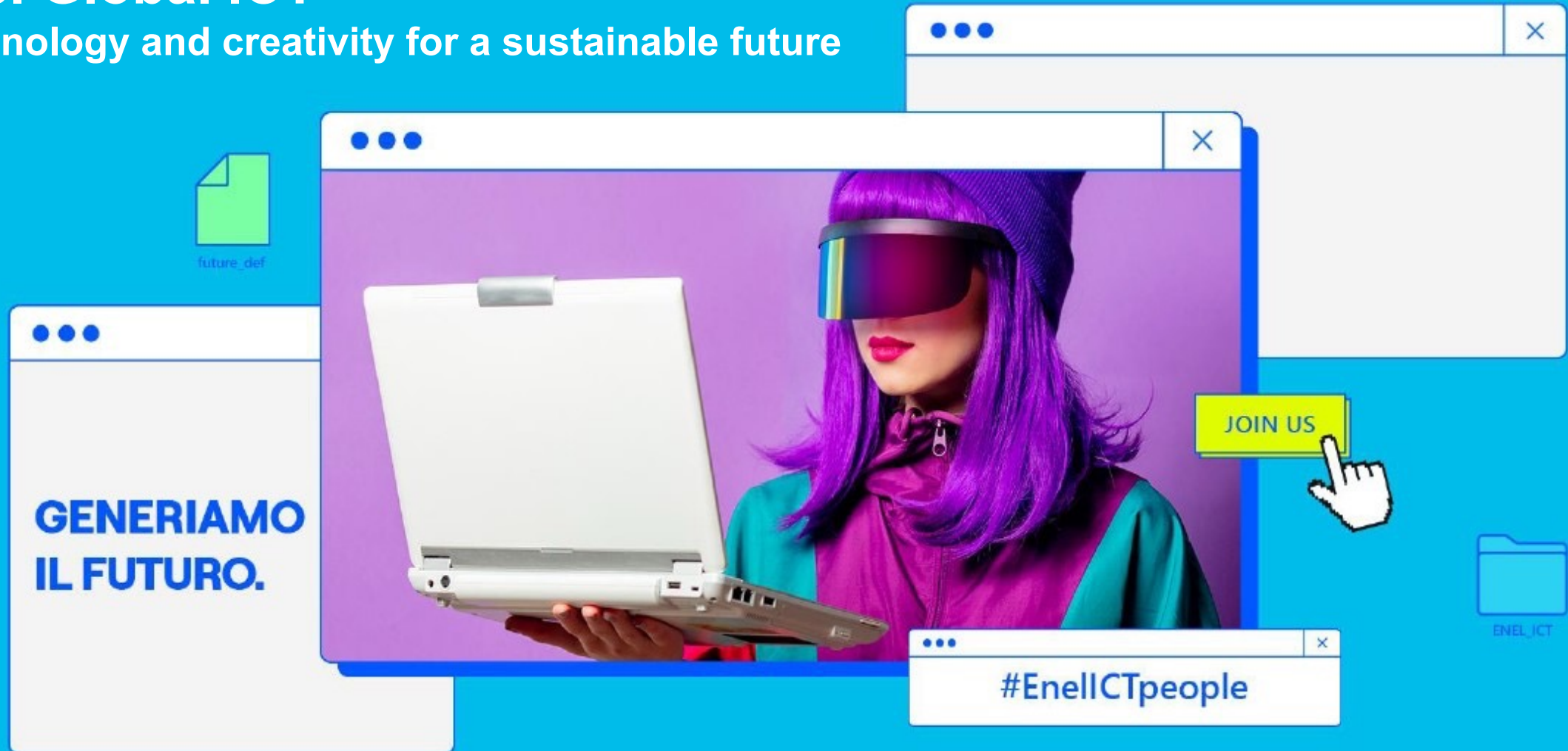


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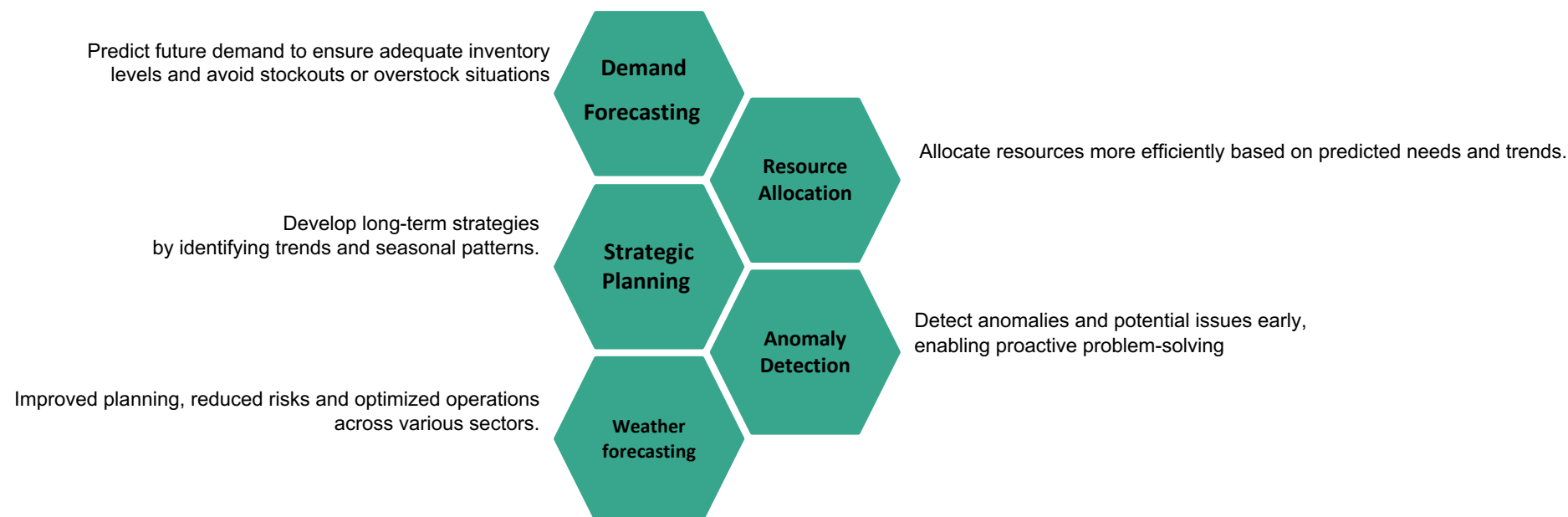


Project Work



Time Series Forecasting: Key Benefits

Time Series Forecasting is a powerful AI technique that has found practical applications in various industries. By analyzing and interpreting data collected over time, businesses can gain valuable insights into trends, patterns, and future predictions.



Project Work

Why Energy Forecasting Matters in an AI-Driven World



Energy forecasting has emerged as a cornerstone for modern energy management, enabling power systems to efficiently balance supply and demand.

In today's world, energy consumption is volatile and influenced by multiple factors — ranging from weather to economic activity. With accurate energy forecasting, we can:

- **Reduce Costs:** Forecasting helps utility companies optimize resource allocation and reduce wasteful overproduction.
- **Enhance Stability:** Predictive AI enables grid stability, avoiding blackouts and equipment strain due to fluctuating demands.
- **Promote Sustainability:** Accurate demand predictions make it easier to incorporate renewable energy sources effectively.

Project Work

Objective



Time Series Modeling in Electric Load Forecasting

The goal of this project is to develop a Time Series Forecasting model capable of predicting future electricity consumption based on historical patterns.

1. Conduct Exploratory Data Analysis (EDA) for temporal data (seasonality, trends, noise)
2. Implement and compare at least two of the following approaches:
 - a) classical statistical models (e.g. ARIMA/SARIMA)
 - b) deep learning models (e.g. MLP/LSTM/GRU)
3. Evaluate model performance using standard metrics like Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE)

Project Work

Dataset



Source: [Individual Household Electric Power Consumption dataset from the UCI Machine Learning Repository.](#)

Description: contains 2,075,259 measurements gathered in a house located in Sceaux (7km of Paris, France) between December 2006 and November 2010 (47 months).

Key Variables:

- Global_active_power: Total active power consumed by the household (kW).
- Voltage: Minute-averaged voltage (V).
- Global_intensity: Minute-averaged current intensity (amp).
- Sub_metering_1, 2, 3: Energy consumption for specific areas (kitchen, laundry, climate control).

Project Work

Deliverables



- **Technical implementation (source code)**
 - A fully documented notebook (.ipynb)
 - A requirement.txt file listing all necessary libraries to ensure reproducibility.
- **Technical report (max 5 page .pdf file)** containing: the EDA summary, the model architecture (including parameters), a results table comparing models using RMSE and MAE, performance visualizations (comparative plots such as “Actual vs Predicted”, residual plots...)
- **Executive summary (a brief .ppt file consisting of 1-2 high-impact slides)**

Thank you

For your attention!